

Working with NGS Primary Data: from Raw Reads to Interpretation

Q to answer in the report: What was the initial quality of the reads, how did the trimming improve it? Specify the trimming parameters. Provide the screenshot of MultiQC overview of the quality of initial reads and after the trimming in one figure.

Before trimming a lot of reads have per base sequence quality below 20, especially closer to the end. Also TruSeq Adapter sequence is overexpressed. Trimmomatic improved mean quality slightly, but mostly reduced the variability, by removing outliers with bad quality scores.

Trimming parameters:

```
java -jar /home/Shared/tools/Trimmomatic-0.39/trimmomatic-0.39.jar PE -phred33  
/home/Shared/data/SARS/SRR12278429_1.fastq  
/home/Shared/data/SARS/SRR12278429_2.fastq -baseout trimmed.fastq  
ILLUMINACLIP:/home/Shared/tools/Trimmomatic-0.39/adapters/TruSeq3-PE.fa:3:30:1  
0 LEADING:20 TRAILING:20 SLIDINGWINDOW:10:15 MINLEN:60
```



In this MultiQC picture the top lines correspond to trimmed reads, the bottom lines – to initial reads.

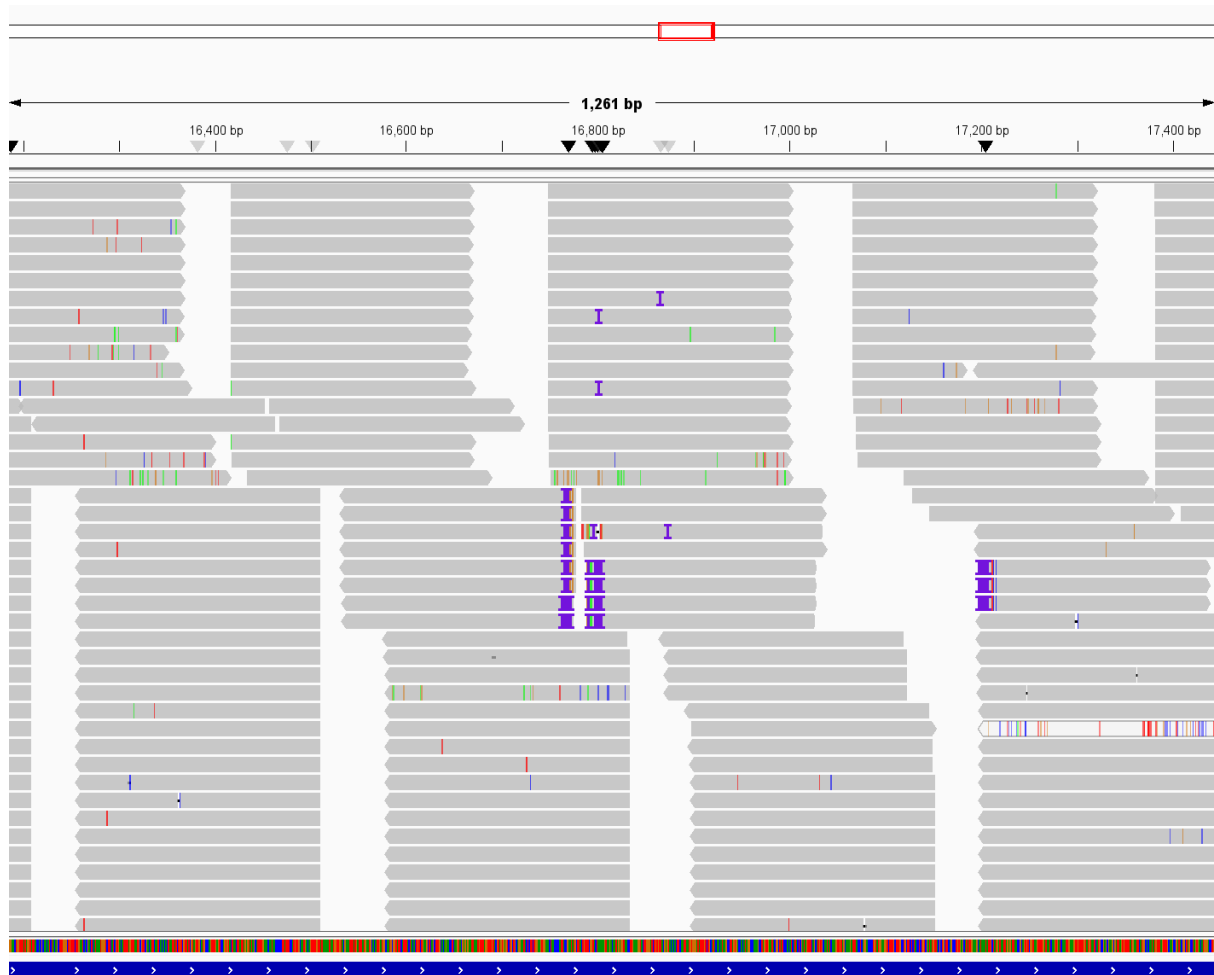
Q to answer in the report: What was the percentage of aligned reads?

95.91% overall alignment rate

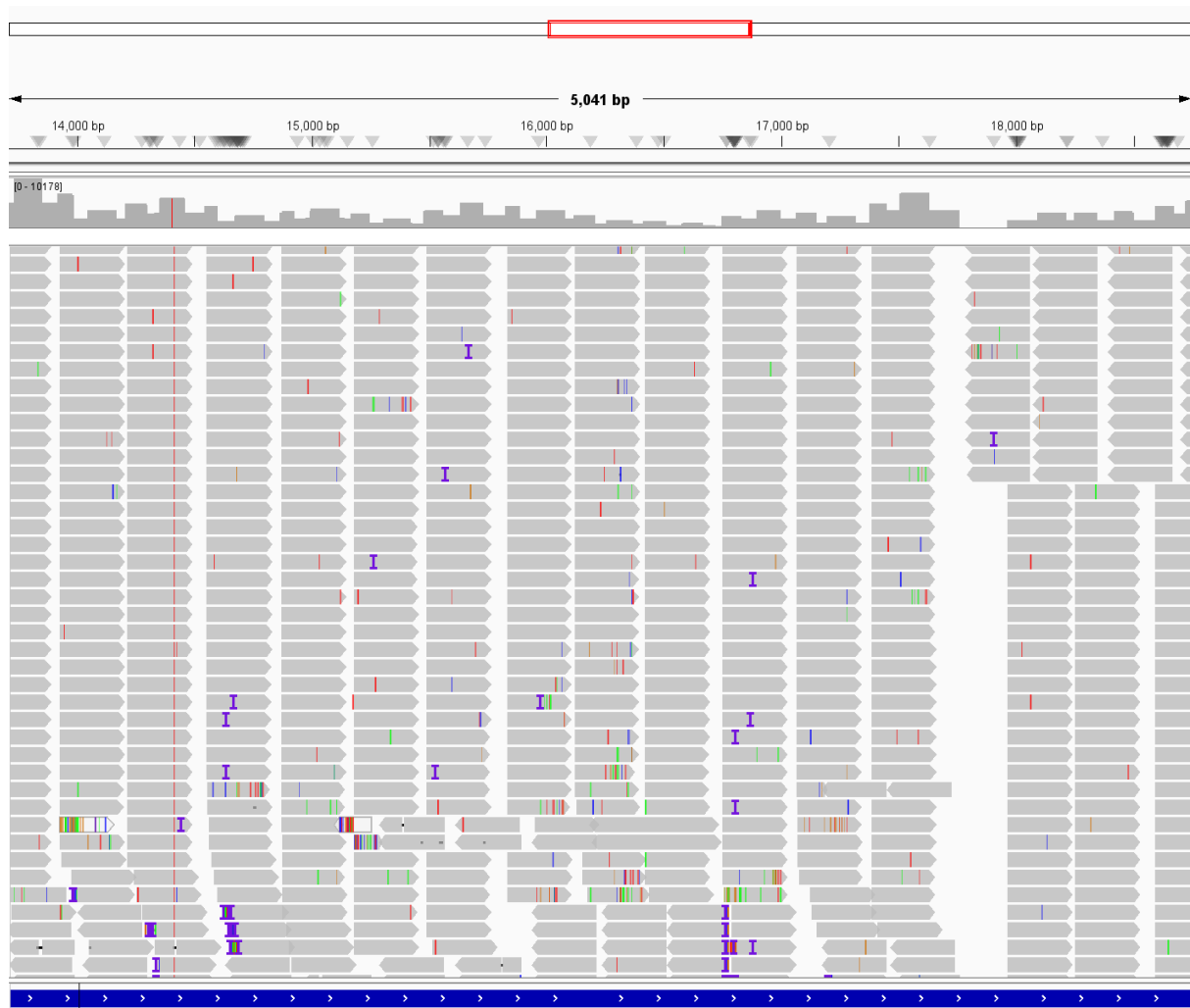
Q to answer in the report: What was the average coverage?

average coverage $Av = 2200.07$

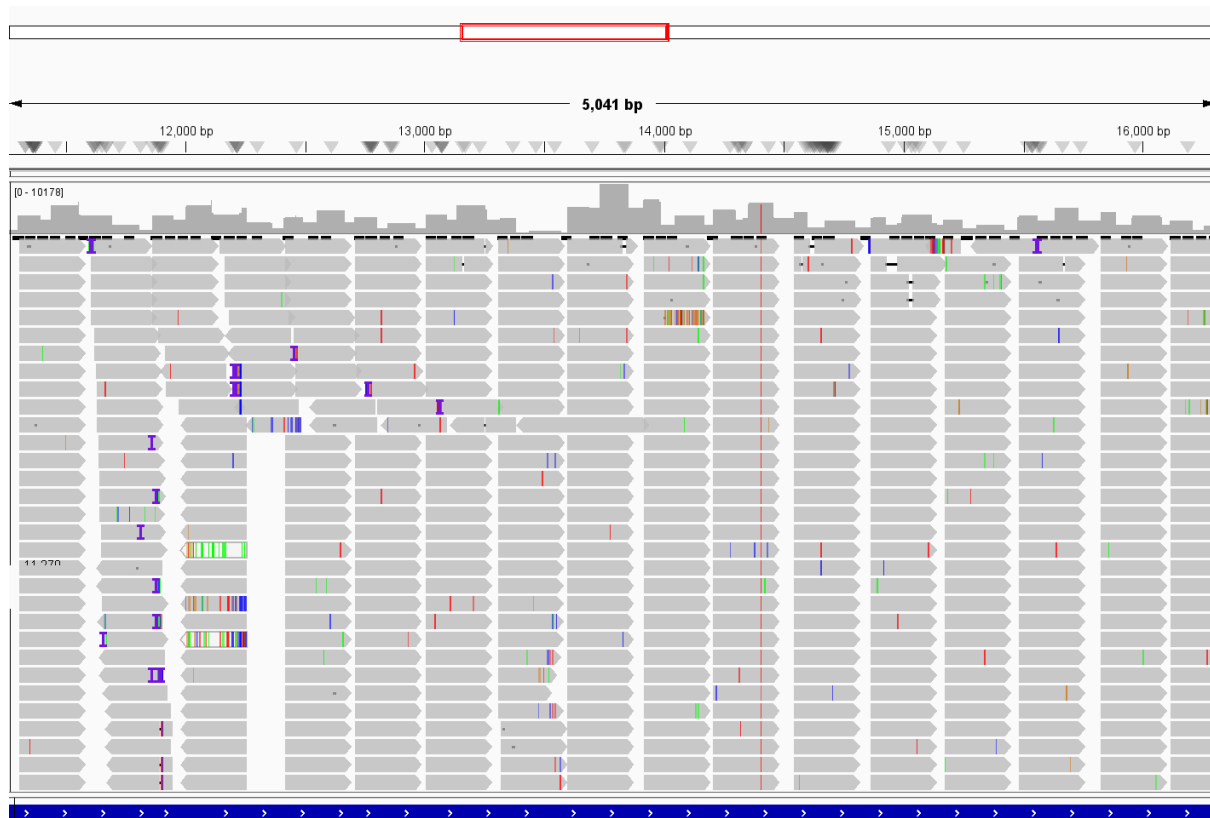
Q to answer in the report: Find and zoom 3 interesting from the biological or technical point of view (huge amount of mutations, high conservativity, no coverage, maximum coverage, etc.) regions of the viral genome? Provide the screenshots in IGV and their description of why these arias are featured.



In this picture there are several reads shifted compared to most reads, i.e. to the position of the primer, but they are still full-length, around 250bp reads. And they have identical insertions at the 5'. That means that this insertion happened before illumina sequencing at PCR amplification stage, so that several such chimeric reads got produced and then sequenced.



In this picture closer to the right edge there is a region with low coverage. This region is exactly the size of a single read. That probably means that something happened at the site of promoter binding, or something happened with the promoter itself, so that there were almost no amplicons produced during the PCR stage and there was nothing to sequence.



In this picture there is a very high coverage region in the middle. In this case it looks like two adjacent amplicons overlap on top of having good primers and good GC content. It is the likely case, because the coverage at that position is roughly doubled compared to other high coverage regions, which probably also happened to have good promoters. This particular one is twice higher than them because two amplicon regions overlap here.

Q to answer in the report: What do the `--min-var-freq` parameter control? What could be achieved by changing the parameter (you may think in terms of the quality of the variant prediction and a number of false-positive and -negative predictions)? Calculate the number of mutations per kb.

`--min-var-freq` sets the minimum fraction of reads supporting the variant. Higher value calls more conservative variants. Lower value increases sensitivity to minor variants but risks calling more sequencing errors. Increasing the value causes fewer false positives but more false negatives, especially for low-frequency variants.

10 SNP variant positions were reported.

Mutations per kilobase = $10/29903 \times 1000 = 0.3344$